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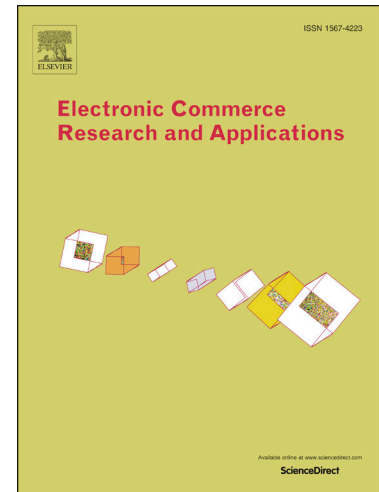
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Are Search Engines Biased? Detecting and Reducing Bias using Meta Search Engines

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Abstract

The *search neutrality debate* stems from content or service providers complaining about being discriminated and ranked unfairly low by search engines, raising the need for methodologies and tools to verify bias in search engine rankings.

For that purpose, we propose in this paper a simple yet effective framework based on the comparison of the results provided by several search engines, and build the corresponding tool to carry out a campaign of tests.

The main objectives are to develop an interpretable model of search engine behaviors and to design statistical tests pointing out suspicious instances as possible bias, without knowing the detailed ranking algorithms implemented by search engines. Our approach consists in reasoning in terms of the *visibility* that search engines give webpages when ranking them among their results; different types of possible bias can then be detected using statistical test of outlier detection.

We apply this methodology to a test campaign over the most searched terms, which highlights some similarities and discrepancies among search engines, and possible instances of bias. Our approach can be of interest to regulators or any actor in the Internet, and is directly applicable to any search term through a publicly-available tool performing extensive comparisons and bias investigations, and offering two (bias-reducing) meta rankings.

Keywords: search neutrality, search bias, outlier detection, meta ranking

1. Introduction

1.1. Context: the search neutrality debate

3 When we want to reach some content, information or tool on our smart-
 4 phone, tablet or laptop, and do not know exactly what to look for and where
 5 to get it, we usually use a *search engine* (SE), which responds to typed-in
 6 search terms with a ranked list of links to the supposed most relevant web-
 7 pages. Google is the most used SE worldwide, but many others exist such as
 8 Bing, Yahoo, Duckduckgo, Qwant, or Startpage. For a more comprehensive
 9 list, see for example [https://en.wikipedia.org/wiki/List_of_search_](https://en.wikipedia.org/wiki/List_of_search_engines)
 10 [engines](https://en.wikipedia.org/wiki/List_of_search_engines). Search engines propositions have a huge influence on our daily life,
 11 since users have a tendency to click on one of the first proposed links, and
 12 search engines are extensively used: Google alone registers in a year about
 13 two trillions searches all over the world¹.

14 But SEs are accused of voluntarily lowering some links in their rankings,
 15 hence reducing the visibility, access, and revenue of the corresponding actors,
 16 or of boosting other links the SEs are closely or financially related to². While
 17 SEs are making money thanks to *sponsored links*, that are displayed links
 18 clearly presented as advertisements corresponding to the queries and sold
 19 through auctions [1, 2], we are talking here about *organic links*, representing
 20 the considered most relevant responses to the dialed search terms. This
 21 issue of biasing the results was brought in 2009 by Adam Raff, co-founder of
 22 the price-comparison company Foundem, saying that Google was voluntarily
 23 penalizing his company in rankings with respect to Google's own services.
 24 Any SE could indeed in theory choose which company to promote or to
 25 demote with financial motivations; such an optimal strategy was computed
 26 in [3], as a trade-off between short-term and long-term revenues; other models
 27 can be found in [4].

28 This has led to the so-called and still vivid *search neutrality debate*, ask-
 29 ing whether SEs should be neutral in their organic rankings—i.e., base their
 30 rankings on relevance only—and if so, how that can be monitored [5, 6, 7, 8].
 31 The issue is related to the sensitive *Net neutrality debate* [9, 10, 11, 12, 13],
 32 active for almost 20 years, asking whether Internet Service Providers (ISPs)
 33 should be allowed to differentiate service at the packet level within the net-
 34 work to get some return on their infrastructure investment. A higher quality
 35 of service would be paid by content providers, but this raised concerns re-

¹<https://searchengineland.com/google%2Dnow%2Dhandles%2D2%2D999%2Dtrillion%2Dsearches%2Dper%2Dyear%2D250247>

²See for example <https://www.technologyreview.com/s/610275/meet%2Dthe%2Dwoman%2Dwho%2Dsearches%2Dout%2Dsearch%2Dengines%2Dbias%2Dagainst%2Dwomen%2Dand%2Dminorities/>

garding the impact on innovation, freedom of speech (if some content can be blocked), etc. The basic definition of Net neutrality, even if it depends on countries, is that ISPs should treat equally all content flowing through their networks, regardless of its kind, source, or destination. But if the key issue is access and impact on innovation, shouldn't SEs be part of the equation? Indeed, similarly to ISPs, search engines have a *gatekeeper* position and can significantly distort the competition among actors on the Internet.

There is reluctance to accept the search neutrality debate in the same way as the Net neutrality debate. First, the nature of an SE is often argued to allow to make various and sometimes new/surprising propositions to open-minded users; but in relation with Net neutrality, why should helping some quality-demanding applications correspondingly be forbidden by ISPs? Second, it is often advocated that if you are not happy with an SE, switching to another is just "one click away". But it might not be easy to detect biases and switch SEs when unfairness is just for a limited number of queries, and the SE market is not as competitive as it seems, with as of June 2020³ Google concentrating 91.75% of worldwide searches, the followers being Bing (2.75%), Yahoo (1.7%), Baidu (1.02%), Yandex (0.92%), and DuckDuckGo (0.49%). Furthermore, the reasons that may lead users to adopt an SE might include aspects other than the result quality (such as eco-friendly measures) [14, 15], hence ensuring a level playing field on each search engine remains important. Finally, it is often thought that defining a neutral behavior is simple for an ISP but not for an SE. But defining packet-level neutrality as "equal treatment at nodes of the network" does not encompass differences at the flow level, which could for example come from different round-trip times when using TCP (dependent on distance), a possible discrimination, and does not deal with activities such as content delivery networks [13]. For SEs, addressing the neutrality debate also means introducing a definition of a neutral behavior in a clear and objective manner. A fair way would be to rank according to *relevance* to queries, but the notion of relevance is difficult to define, subjective, since potentially very user-dependent. Personalization of SE ranking, using location or cookies, refines but also complicates the notion. A "solution" to deal with possible non-neutrality could be to ask SEs to be transparent about the used algorithms, allowing regulators or users to be informed and check whether

³See <http://gs.statcounter.com/search-engine-market-share>

voluntary discrimination is implemented, an approach more and more pushed by (European) regulators⁴. That solution however induces secret/copyright issues; we do not enter those considerations/assumptions in the present paper and rather intend to design a solution robust to the opacity of ranking algorithms.

1.2. Related work

Defining relevance is a very sensitive and complicated issue, the considered definition being revealed by SEs or not. The PageRank method [16] initially used by Google is an example of such an algorithmic definition, basically ranking pages based on their incoming links (backlinks), but is computationally demanding. We will take a simpler alternative here, by making use of already existing ranking systems by SEs (hence no computationally-demanding algorithm to add): we will define the relevance of a page/link as the average *visibility* of its positions among SEs. Given some visibility values of the different positions (e.g., estimated from measurements on the click behavior of users), that metric is simple to compute, and smooths potential irregularities by SEs. The issue of fairness in rankings has also been addressed in [17, 18] but not with the notion of visibility and click-through-rates which specifically concerns search results.

There exist many works on search non-neutrality, with sometimes different conclusions. Mowshowitz and Kawaguchi [19] use an own-defined measure of similarity between SE results, and show that there are significant differences, particularly sensitive to the subject domain but not the search terms in a given domain. The same authors claim similarly to us that “A family of comparable search engines can be used to approximate the ideal or fair distribution of items retrieved for a set of queries” [20], but they do not provide statistical tests to detect bias, and instead use a notion of dissimilarity not involving any interpretation in terms of visibility. A comparison and illustration of differences among SEs is discussed in [21], trying to determine the possible causes. Reference [22] also focuses on comparing ranked lists, including truncated lists with partly different elements; see also [23]. [24] analyses how own content by Bing or Google is ranked first more than on competing engines to highlight a general bias. Our work can also be linked with the study of political bias and the role of search engines in elections

⁴<https://ec.europa.eu/digital-single-market/en/online-platforms-digital-single-market>

[25, 26], but the latter involves different measures related to personalization. Evaluating the performance of SEs (information retrieval) is another related topic [27], the score we define being original in this respect, as it requires no human but is based on the assumption that SEs point out to the right links on average.

1.3. Paper contributions and organization

The purpose of this paper is multifold. Our contributions, and the differences with respect to the literature can be decomposed as follows.

1. We formalize a definition of visibility which will be used to rank pages (links). The score of a page will be the average displayed interest/visibility of the positions obtained by that page at SEs. In this sense the score is applying a weighted Borda rule [28, 29]. As a consequence, we are also able to compute a score for each SE, by summing the scores of the presented pages/links weighted by their positions. We can then easily compare SEs for a given search term and for a population of search terms.
2. We develop a statistical methodology to identify potential bias based on Dixon's Q test. The proposed tests are more precisely: i) Looking at scores of SEs for a search term, an excessively low score being suspect since it indicates a ranking very different from other SEs. ii) Investigating whether the highest-ranked link of a SE is ranked as high by others. iii) Looking at the rankings of the most consensual link and checking whether that link is ranked abnormally low by some SEs. We are not aware of any other work using a similar statistical testing procedure to detect search engine bias.
3. We propose two meta-engines. The first one, called *consensus SE*, ranks links according to the weighted Borda scores (average visibility among SEs) obtained above. The second one is following the *majority judgment* rule defined in [30] for votes and adapted to our context. Voters are SEs voting for pages as candidates. Pages are ranked according to the median of votes, that is the rank for which 50% or more of the SEs give at least this rank. This ranking is expected to erase bias more than the consensus SE and we provide a theoretical framework to illustrate it through simulations. Aggregating search results to get a meta-engine is not new (see [18]), but our two aggregation techniques (*weighted Borda* or *majority judgment*) have not been implemented so far to our knowledge.

- 142 4. We present a publicly-available platform/web page <https://snide.>
143 [irisa.fr/](https://snide.irisa.fr/) on which a search term can be composed, the results of 15
144 engines plus the two meta engines are displayed, and potential bias are
145 pointed out, if detected by statistical tests.
- 146 5. We provide results of bias on searches through statistics of failed tests
147 for each SE, and give illustrative examples.
- 148 6. We compare the meta-engines in terms of average scores, highlight how
149 they smooth bias, and provide statistics on their performance.
- 150 7. Thanks to extensive studies on campaigns of most dialed search terms,
151 we compare the results of all SEs (not only our meta-engines, but also
152 the “regular” ones). We are able to compute and compare expected
153 scores over searches, but also their distributions, in order to see similar-
154 ities, highlight if they use close algorithms, and compute the frequencies
155 of potential bias to emphasize how often they can be trusted. Again,
156 comparing ranking algorithms has already been done in the literature,
157 see [19, 18, 31, 22], with methods mostly coming from models related to
158 voting systems. An example is to use the Kendall distance, looking at
159 the pairwise ordering disagreement between lists. But those methods
160 do not take into account the weights associated to CTRs of positions
161 and their economic impact, nor do they compute distributions of scores
162 and bias.

163 Our model and implemented tool can be used to detecting outliers among
164 the results of several search engines; note that they are not systematically
165 the consequence of a voluntary bias of the search engine and be due to sev-
166 eral reasons, including a different notion of relevance or different user-related
167 information. But the methodology allows to point out where further investi-
168 gation should probably be carried out.

169 The paper is an extension of a conference paper where preliminary results
170 were presented [32]. While the principle of using visibility as a page score
171 and the consensus search engine were already introduced, the additional re-
172 sults are: i) the statistical tests of bias, the main purpose of this activity; ii)
173 an analysis of outlier detection on scores for an extensive list of representa-
174 tive search terms; iii) the introduction of the second meta-engine based on
175 majority judgment theory and believed to further reduce bias with respect
176 to the consensus one; iv) a comparison of all SE outputs as well as our two
177 meta-engines; v) the use of a more complete list of 15 SEs, while “only” 9
178 were considered in [32]; vi) the introduction of a publicly available tool to

179 compare SEs and perform bias detection tests.

180 The rest of the paper is organized as follows. Section 2 presents our
 181 model, more precisely how we compute scores of links and SEs for given
 182 search terms based on average visibility. Section 3 introduces the statistical
 183 tests for potential bias, each of them based on Dixon’s Q test for outlier
 184 detection. Given the several SEs available, Section 4 proposes the two meta-
 185 engines aiming at reducing bias. The whole methodology is implemented
 186 on a publicly available web site to detect and reduce bias (thanks to our
 187 meta engines) as described in Section 5. Extensive tests are conducted and
 188 analyzed in Section 6, on detected bias but also to compare all the search
 189 engine outputs. Finally we conclude and describe our next research directions
 190 in Section 7.

191 2. Model/scoring system

192 This section recalls and expands the discussion in [32].

193 2.1. Score/relevance: visibility among SEs

194 The ranking of an SE is supposed to correspond to a list of links ordered
 195 by their relevance with respect to the composed search term. Defining the
 196 notion of relevance is a complicated issue which can be considered subjective.
 197 Our goal is not to discuss how relevance should be defined, and it seems good
 198 to observe several different notions implemented to leave the choice to (in-
 199 formed) users; this problem is also often related to the notion of transparency
 200 of algorithms but is not the main focus of the present paper (for more, see
 201 [13]). SEs use computationally-demanding algorithms to set their relevance
 202 score, such as PageRank for Google [16]. Bing⁵ as another main SE defines
 203 relevance as

204 “how closely the content on the landing page matches the intent
 205 behind the search query. This includes matching terms directly
 206 on the page as well as terms used in links referring to the page.
 207 Bing also considers semantic equivalents, including synonyms or
 208 abbreviations, which may not be exact matches of the query terms
 209 but are understood to have the same meaning.”

⁵See <https://www.bing.com/webmaster/help/webmaster-guidelines-30fba23a>

210 But Bing also uses other parameters for its ranking, such as the quality/reputation
 211 of the page, the fact that users already clicked on this link, the user location,
 212 the page load time, etc. The algorithm is therefore complex and resource
 213 and experience demanding.

214 We choose to make use of the outputs from the (already existing) complex
 215 algorithms of SEs rather than defining our own (which could be seen at least
 216 as arbitrary as any SE). For a given search term, our definition of a *score*
 217 (or relevance) of a page will be its average visibility over SEs, the visibility
 218 for an SE depending on the page's position in the ranking [1]. That choice
 219 presents several advantages:

- 220 • The method is computationally simple, taking benefit of the complex
 221 algorithms implemented by SEs; we just need to launch a search on all
 222 SEs and associate visibility values to the different positions⁶.
- 223 • What typically counts for content providers is to be clicked, the com-
 224 plaint in the search neutrality debate coming from the lack of reacha-
 225 bility. It therefore makes sense to focus on visibility and quantify the
 226 differences with respect to that notion.
- 227 • Averaging over SEs reduces potential biases in relevance expressions
 228 and/or considers all dimensions of relevance considered by SEs, even if
 229 neglected by others.

230 2.2. Formalization

231 We formalize the model in this subsection, recalling and extending the
 232 notions in [32].

233 We consider n SEs, a set $\mathcal{K}$ of search terms (possibly infinite), and a finite
 234 set $\mathcal{L}$ of pages/links corresponding to all the results displayed for the whole
 235 set of searches. For any search term $k \in \mathcal{K}$, let p_k be its probability to be
 236 dialed during a search, such that $\sum_{k \in \mathcal{K}} p_k = 1$; p_k is therefore related to the
 237 frequency of search term k .

238 Even if SEs can display several pages of results for a given search we will
 239 limit ourselves to the first displayed result page for each SE, and assume for

⁶Those visibility values (at least relative values, which is sufficient for us) can be es-
 timated from the click-through-rate (CTR) values on the links displayed in different po-
 sitions; we keep that estimation out of the scope of this paper and assume the visibility
 values are given.

240 simplicity that it shows a number $a > 0$ of results, even if it can be easily
 241 generalized to the heterogeneous case.

As explained above, we will define the score of a page/link as its average visibility among SEs, a notion easier to consider than the subjective and many-parameters-dependent notion of relevance. Visibility can be estimated through the *click-through-rate* (CTR) of a page at a given position, measuring the rate, or probability, at which it is clicked, denoted by $c_{i,\ell}$ for a Page i at position ℓ [1]. That value is typically SE-, position- and link- dependent, but we assume here for convenience, and as commonly adopted in the literature, a separability property [1] formalized as

$$c_{i,\ell} = q'_i q_\ell,$$

242 i.e., the CTR is the product of a link-specific factor q'_i (some links are more
 243 attractive by nature) and a position-specific factor q_ℓ for position ℓ . We
 244 will interpret that expression in terms of probabilities, with the CTR $c_{i,\ell}$
 245 (probability of being clicked) being the product of a *probability of being seen*
 246 q_ℓ (that depends on the link position) and a *probability of being clicked when*
 247 *seen* q'_i (that depends on the link's relevance to the request). We will therefore
 248 identify the visibility that a SE gives to a link/page shown in position ℓ with
 249 the term q_ℓ . In general those terms can also be SE-dependent, but we will
 250 here consider the same values across SEs, since estimating them would require
 251 data obtained from extensive experiments or directly from SEs. Nevertheless,
 252 that aspect is generalizable in our model without technical difficulty. Finally,
 253 note that we typically have $q_1 \geq q_2 \geq \dots \geq q_a$, meaning that the better
 254 ranked, the more visible.

255 The visibility values q_ℓ we will use in our numerical studies come from
 256 CTR values obtained through measurements in [33] and displayed in Table 1,
 257 with $a = 10$. Note that only the relative values of (q_ℓ) are needed, multiplying
 258 all of them by a positive constant does not change anything in the model
 259 or the results. The underlying assumption for obtaining those values from
 260 statistics on click numbers is that the link-specific terms will tend to average
 261 out over a large number of searches, so that the click probabilities on each
 262 position will be proportional to the position-specific terms (q_ℓ) . Note also
 263 that it is easy to change those values⁷ and run the methodology proposed in

⁷One may for example use the click-through-rate estimations obtained more recently, available at <https://www.sistris.com/blog/why-almost-everything-you-knew-about-google-ctr-is-no-longer-valid/>

Table 1: CTR values used in the paper, taken from [33]

q_1	q_2	q_3	q_4	q_5	q_6	q_7	q_8	q_9	q_{10}
0.364	0.125	0.095	0.079	0.061	0.041	0.038	0.035	0.03	0.022

264 this paper.

265 Having introduced all the required notations, we can define the score of
 266 a page for a given search term from outputs provided by SEs.

267 **Definition 1.** *The score $R_{i,k}$ of a page $i \in \mathcal{L}$ for a search term $k \in \mathcal{K}$*
 268 *is defined as the average visibility of that page over the rankings from all*
 269 *considered SEs for that search term:*

$$R_{i,k} := \frac{1}{n} \sum_{j=1}^n q_{\pi_j(i,k)} \quad (1)$$

270 where $\pi_j(i,k)$ denotes the position of page i on SE j for search term k . If a
 271 page is not displayed by an SE, the visibility is taken as 0 (equivalently, we
 272 can define a position $a+1$ for non-displayed pages, with $q_{a+1} = 0$).

273 With that score definition, any bias from an SE is smoothed (reduced)
 274 due to the averaging procedure, so that we expect page scores to be closely
 275 linked to the actual relevance of pages with respect to search terms.

276 2.3. Search engine score

277 Even if it is not the initial purpose of the work, defining a score for pages
 278 allows to define a corresponding score for SEs, for a given search term but
 279 also for the whole set of searches. It quantifies how “consensual” with respect
 280 to the other SEs a given SE is.

281 **Definition 2.** *The score $S_{j,k}$ of an SE j for a given search term k is defined*
 282 *as the total “page score visibility” of its results for that search term:*

$$S_{j,k} := \sum_{\text{page } i \in \mathcal{L}} q_{\pi_j(i,k)} R_{i,k},$$

283 where again $q_p = 0$ if a page is ranked at position $p \geq a+1$ (i.e., not shown).

Note that equivalently, the score can be reformulated by summing over the displayed pages,

$$S_{j,k} = \sum_{p=1}^a q_p R_{\tilde{\pi}_j(p,k),k}$$

where $\tilde{\pi}_j(p, k)$ is the page ranked at the p^{th} position by SE j for search term k , i.e., $\tilde{\pi}_j(\cdot, k)$ is the inverse permutation of $\pi_j(\cdot, k)$.

To interpret that score, the higher an SE ranks highly exposed pages, the higher its score, meaning that the score measures the visibility that the SE gives to pages that are well-exposed on average by SEs. Another way to interpret it is to see the page scores as approximations of the actual page relevances, so the SE score somehow measures the relevance seen by users on that SE.

From the SE scores per search term, we can also define an overall score of SE j over the whole set of search terms:

Definition 3.

$$S_j := \mathbb{E}_{\mathcal{K}}[S_{j,k}] = \sum_{k \in \mathcal{K}} p_k S_{j,k}, \quad (2)$$

where $\mathbb{E}_{\mathcal{K}}[\cdot]$ is for the expected value of probability distribution $(p_k)_{k \in \mathcal{K}}$ over $\mathcal{K}$.

This definition takes into account the frequency/probability p_k of each search term $k \in \mathcal{K}$, so that S_j represents the expected score that SE j would obtain when a (random) search term is typed in.

3. Statistical tests of bias

Our main goal is to develop a tool to detect potential biases in SEs' outputs, based on the lists of results from all SEs. This is typically the goal of *outlier detection* [34, 35]. We want to use a *non-heuristic* statistical test designed for a small to moderate sample size (the size being the number of SEs): the natural candidate is then Dixon's Q test [36, 37].

3.1. Dixon's Q test

Dixon's Q test [36, 37] is designed to quickly identify outliers in datasets that only contain a small number of observations.

Assume a data set of n (independent and identically distributed) values normally distributed and sorted in increasing order $x_1 \leq x_2 \leq \dots \leq x_n$.

310 To see if the largest value is an outlier, define the statistics

$$Q := \begin{cases} Q_{10} = \frac{x_n - x_{n-1}}{x_n - x_1} & \text{if } 3 \leq n \leq 7 \\ Q_{11} = \frac{x_n - x_{n-1}}{x_n - x_1} & \text{if } 8 \leq n \leq 10 \\ Q_{21} = \frac{x_n - x_2}{x_n - x_{n-2}} & \text{if } 11 \leq n \leq 13 \\ Q_{22} = \frac{x_n - x_2}{x_n - x_3} & \text{if } 14 \leq n \leq 25. \end{cases} \quad (3)$$

311 Values $q_{\alpha,n}$ are tabulated (see [37]) such that x_n is considered an outlier
312 at confidence level $1 - \alpha$ (or risk α) if $Q > q_{\alpha,n}$.

313 Similarly, if we rather want to investigate whether the smallest value x_1
314 is an outlier, we will consider the statistics

$$Q := \begin{cases} Q_{10} = \frac{x_2 - x_1}{x_n - x_1} & \text{if } 3 \leq n \leq 7 \\ Q_{11} = \frac{x_2 - x_1}{x_n - x_1} & \text{if } 8 \leq n \leq 10 \\ Q_{21} = \frac{x_3 - x_1}{x_{n-1} - x_1} & \text{if } 11 \leq n \leq 13 \\ Q_{22} = \frac{x_3 - x_1}{x_{n-2} - x_1} & \text{if } 14 \leq n \leq 25. \end{cases} \quad (4)$$

315 The Gaussian assumption is often violated, which may be a problem for a
316 small data set, and it is also often advised to use Dixon's Q test to only detect
317 outliers and not to remove them⁸, but it is the existing tool that corresponds
318 best to our purpose, namely statistically detect potential bias/outlier.

319 We will consider $n = 15$ SEs, that is, we will use the statistics $Q = Q_{22}$.

320 The next four subsections describe the four implementations of Dixon's
321 Q test we will consider in practice.

322 3.2. Abnormal SE score for a given search term

323 The first test we will consider aims to determine whether, for a given
324 search term, an SE score as of Definition 2 is abnormally low with respect to
325 the others.

326 For a search term $k \in \mathcal{K}$, the values used for the test are therefore $x_j =$
327 $S_{(j),k}$, the SE scores, sorted in increasing order. We consider the statistics Q
328 defined in (4), and will say that the SE with the smallest score is an outlier
329 and potentially biased at risk α if $Q > q_{\alpha,n}$.

330 By that test, we identify SEs with outputs significantly different from
331 competitors. An uncommon behavior with respect to competitors means a
332 low score.

⁸See https://sebastianraschka.com/Articles/2014_dixon_test.html

3.3. Investigating SEs disregarding the most visible link

Consider the page/link ranked as the most visible for a given search term k (i.e., the page with the largest score $R_{\cdot,k}$). Our aim is to determine if an SE is ranking and making abnormally less visible this page considered relevant by the consensus.

If ℓ^* is the page in $\mathcal{L}$ with the highest score, the values x_i for $1 \leq i \leq n$ are the ordered scores/visibilities among the n SEs, that is, $x_i = q_{\pi_i(\ell^*,k)}, 1 \leq i \leq n$.

We again consider the statistics Q defined in (4), and will say that the SE with the smallest score is an outlier and abnormally “hides” the most relevant page (as per the consensus ranking) at risk α if $Q > q_{\alpha,n}$.

3.4. Investigating if the top-ranked page of each SE is also visible at other SEs

The third test is to investigate whether an SE abusively gives the top spot to a link (as we did in our simulations in Subsection 4.2.2) to create an abnormal advantage.

To do that for any search term $k \in \mathcal{K}$, consider for each SE j the top-ranked page $\ell'(j)$, and then take $(x_i)_{1 \leq i \leq n}$ as the ordered visibilities $(q_{\pi_i(\ell'(j),k)})_{1 \leq i \leq n}$ of that page among SEs.

We end up with n tests for $1 \leq j \leq n$ suggesting that at risk α the visibility of the top-ranked page at SE j (necessarily the largest visibility q_1) is abusively high with respect to the visibility of that page at other SEs if the statistics Q defined in (3) verifies $Q > q_{\alpha,n}$.

3.5. Investigating if an SE ranks first a page not considered relevant by others

The last test looks at the visibilities of the pages ranked first by SEs. If one of these top-ranked pages has a visibility (a score $R_{\cdot,k}$) abnormally low with respect to the others, it may mean that this page is biased upward by the SE.

Formally, for a search term $k \in \mathcal{K}$, the values used for the test are therefore $x_j = R_{\tilde{\pi}_{(j)}(1,k),k}, 1 \leq j \leq n$, the ordered n scores of the SE top-ranked pages. We consider the statistics Q defined in (4), and will say that the SE whose top-ranked page has the smallest score is an outlier and potentially biased at risk α if $Q > q_{\alpha,n}$.

366 4. Meta search engines

367 Defining page scores and looking for an SE without bias leads naturally
 368 to the definition of meta SEs with bias reduction. In Section 4.1, we redefine
 369 the SE first introduced in [32], and in Section 4.2 we introduce another meta
 370 SE, based on the majority judgment rule and expected to limit the potential
 371 biases even further.

372 4.1. Consensus SE

373 Using the score definition of pages in the previous section, it seems natural
 374 to define a *consensus SE* as the one showing the pages with the highest scores,
 375 i.e., maximizing the SE score for each search term.

Definition 4. *For a given search term k , the consensus SE is a meta-SE that shows at position p the page $\pi^{(k)}(p)$, where the permutation $\pi^{(k)}(\cdot)$ is chosen such that*

$$\pi^{(k)}(\cdot) \in \arg \max_{\pi(\cdot)} \sum_{p=1}^a q_p R_{\pi(p),k}.$$

376 *In other terms, pages are displayed in decreasing order of their computed*
 377 *score (their “estimated relevance”) $R_{\cdot,k}$ for that search term k .*

378 We will also be able to compute the total score of the consensus SE,
 379 maximal among all SE scores by definition, which will allow us to compare
 380 how close to “optimal” the other SEs are.

381 4.2. Majority judgment SE

382 4.2.1. Definition

383 Another meta SE we propose here is based on the so-called majority
 384 judgment considered for voting systems and presenting [desirable](#) properties
 385 [30, 38]. We adapt it to our context to build a ranking procedure expected
 386 to further reduce biases induced by SEs with respect to the consensus SE.

387 Consider a search term $k \in \mathcal{K}$. In the majority judgment, “voters” give
 388 an evaluation to each candidate. In our context, voters are SEs, and candi-
 389 dates are pages to be displayed. The evaluations are the visibility values (or
 390 equivalently the positions) associated to each page, $r_{i,k,j} := q_{\pi_j(i,k)}$ for page i
 391 on SE j : pages ranked first have the highest scores/evaluations.

392 **Definition 5.** *The majority judgment ranking for a search term k works as*
 393 *follows:*

- For each page i , compute its majority judgment $r_{i,k}$ as the best (highest) evaluation among $(r_{i,k,j})_{1 \leq j \leq n}$ for which 50% or more SE give at least that value. In case of an odd number n of SEs, it is the middle value (median) of the ordered values, but for an even number, it is the lowest value of the two in the middle interval (so that more than 50% of the SEs give at least this value). Formally, if $r_{i,k,(1)} \geq r_{i,k,(2)} \geq \dots \geq r_{i,k,(n)}$ are the ordered values,

$$r_{i,k} = r_{i,k,(\lceil (n+1)/2 \rceil)}$$

394 In summary, it is the best score accepted by a majority of SEs.

- The final majority judgment ranking is the ranking according to the values $r_{i,k}$, from the largest to the smallest.

397 In case of equality in a given subset, a vote equal to $r_{i,k}$ is removed for all
 398 competitors i , and the algorithm repeated with this excluded value (for this
 399 subset of competitors) by computing the new majority judgment values. This
 400 is repeated up to a differentiation is made, which happens except if some
 401 candidates have exactly the same set of rankings.

402 Remark that while the consensus ranking allowed naturally to give a score
 403 to SEs and make comparisons (computing through the score how far from the
 404 consensus a given SE is), it is not that obvious with the majority judgment.

405 4.2.2. Sensitivity to bias/outliers: Majority Judgment versus Consensus

406 If an SE is biased and associates to a page a visibility value that is un-
 407 correlated with its relevance, the value itself does not influence the majority
 408 judgment ranking (only its position with respect to the median can have an
 409 impact); that should smooth biases even more than the consensus ranking.

410 To illustrate this, we build an artificial model made of 15 SEs, and 20
 411 potential pages to be displayed. When we simulate a search, each page has an
 412 intrinsic relevance, that we draw from a uniform distribution over the interval
 413 $[0, 1]$. SEs on the other hand may only be able to imperfectly estimate that
 414 relevance: to model that we add for each SE and for each page a Gaussian
 415 random variable with mean 0 and variance σ^2 (all those 15×20 “noise”
 416 variables being independent). If unbiased, SEs then rank pages according to
 417 that estimation, so that the larger σ^2 the more differences between SEs, and
 418 all SEs give exactly the same ranking when $\sigma = 0$.

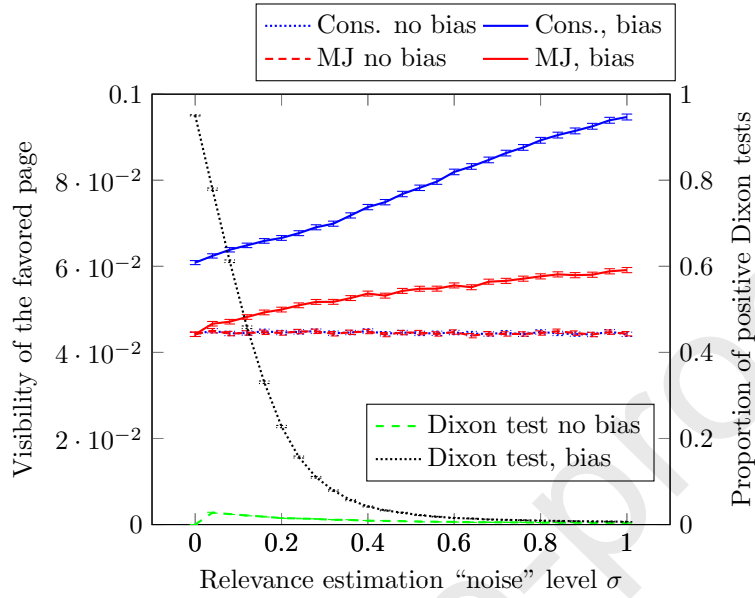


Figure 1: Visibility with the two meta-SEs, with and without bias introduced by the first SE, and corresponding results of the Dixon test for that SE (Dixon test on the SE score being abnormally low). 95% confidence intervals are also displayed.

419 We aim to investigate the impact on meta-SE rankings of one SE biasing
 420 in favor of one page, displaying it first whatever its relevance estimation. (We
 421 assume that the SE still ranks all the other pages by estimated relevance,
 422 and all the other SEs also rank with respect to their estimated relevances.)
 423 We plot in Figure 1, when the “estimation imperfection” σ varies, the aver-
 424 age visibility (along with the corresponding 95% confidence interval) of the
 425 favored page when repeating 10^5 independent experiments for the simulated
 426 and estimated relevances. We use the visibility values (q_ℓ) given in Table 1.

427 We can readily check that average visibilities for the two meta SEs are
 428 similar in value when not biasing, and not sensitive to the noise level σ
 429 in relevance evaluation by SEs. On the other hand, there are significant dif-
 430 ferences in the visibility of biased pages for our meta SEs: i) the bias (i.e.,
 431 the difference of visibility with respect to the un-biased case) is significantly
 432 lower when using majority judgment with respect to consensus; ii) in both
 433 cases, the more variability in relevance estimation, the larger the bias; this
 434 is due to less consensually defined positions (and scores) when variability is
 435 high, hence less consensus among non-biased SEs which leaves more room

to affect meta-SEs; iii) the bias increase with σ is significantly lower for the majority judgment, even close to marginal, as expected. On the same figure, we superimpose the frequency of rejection of the null hypothesis “*the score of SE1 (the one with a potential bias) is not abnormally low with respect to the other SEs*”, with the Dixon test at risk level 1% (described in Section 3.2). Notice that when that SE does not bias the results, it almost never is labeled as an outlier, i.e., false positives are rare. When implemented, bias is frequently detected for low noise values σ : in those cases there is a clear consensus among all the other SEs and the bias is easily detectable. For large noise values, SEs have difficulty agreeing on what pages to display, which benefits the biasing SE in two ways: it has a larger impact on the visibility that its target page gets with the meta-engines, but the bias introduced is harder to detect.

5. A publicly available platform: SNIDE

We have designed a tool/website to implement the theory developed in previous sections. The tool, available at

<https://snide.irisa.fr/>

is both designed for single (general audience) and multiple (research) search purposes⁹. Figure 2 (*left*) displays the home page for single searches (e.g., for a user wishing to use our meta-engines or just compare SEs visually), and (*right*) the page after clicking on the “advanced” button, designed to run the tool for multiple searches (e.g., to carry out larger-scale analyses on the differences among SEs).

For both single or multiple searches, the tool is a web crawler written in Python. The code to get page URLs is adapted to each SE, because they display the results differently. It gives the results from 13 search engines¹⁰ selected among the most used worldwide. Note that we excluded SEs such as Baidu due to language issues. The 13 SEs are:

⁹We continuously update the website to follow the changes implemented by SEs when collecting results; as a result, the website may experience some small issues. When we carried out the analysis presented in this paper, we checked that the result collection was working properly.

¹⁰Our initial web crawler [32] was limited to 9 SEs and was not available as a public web tool

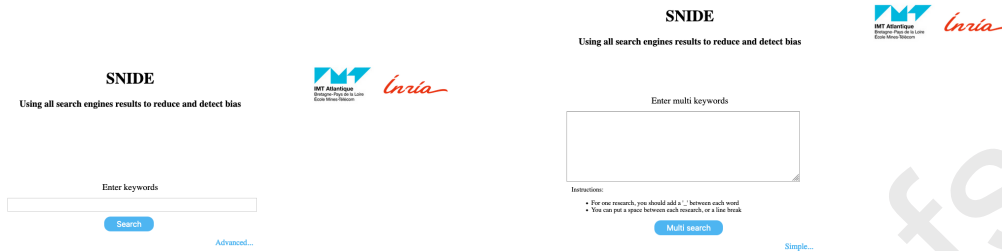


Figure 2: Home page of the website <https://snide.irisa.fr/> (left), and for multiple searches (right) when clicking on “advanced”

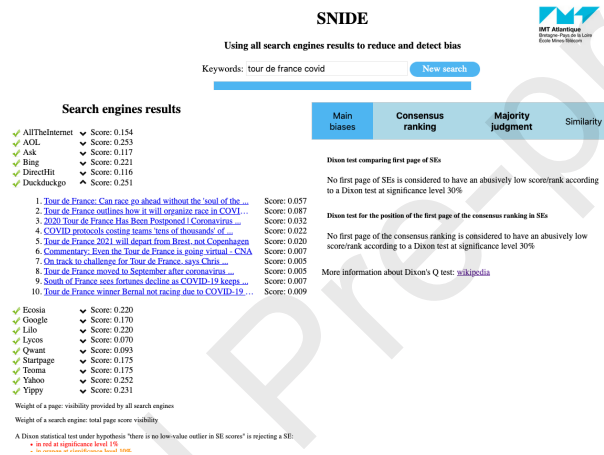


Figure 3: Example of results for a simple search

- AllTheInternet
- AOL
- Ask
- Bing
- DirectHit
- Duckduckgo
- Ecosia
- Google
- Lilo
- Lycos
- Teoma
- Yahoo
- Yippy

464

465 Let us focus on the “simple” search first. The web site provides various
 466 data when composing search terms, as illustrated in Figure 3. The list of SEs
 467 appears on the left, and their scores for this search are displayed. The tool
 468 also highlights in color and indicates below the list of SEs whether one SE
 469 has a score abnormally low according to the test described in Section 3.2. To
 470 go more into details, the user can get the list of the 10 first results of each SE
 471 (together with the scores of those pages) by a click on the “down” arrow near
 472 each SE. On the right-hand side of the page, the tool gives by default the

473 results of the bias tests described in Sections 3.3 and 3.4, and the rankings of
 474 our two meta SEs can be obtained by clicking on the “consensus ranking” or
 475 “majority judgment” buttons. That is useful to make comparisons between
 476 SEs, but may also be of interest for users interested in engines smoothing
 477 possible biases.

478 With the “advanced” option of multiple searches, designed more for re-
 479 search purposes, the user can download two files: one with the full output
 480 (the top 10 results by each SE for each search), and the other including the
 481 analysis (scores and test results). Two format options are possible: text to
 482 immediately visualize the results, or JSON for a use with databases.

483 As often in such a situation, we had to deal with many practical issues.
 484 To list a few:

- 485 • *A priori* different URLs can lead to the same page¹¹ output when ac-
 486 cessing the links proposed by the SEs. Sometimes pages are indiffer-
 487 ently http or https. We have tried to handle these cases as much as
 488 possible.
- 489 • For some specific search terms such as *Facebook*, some SEs do not pro-
 490 vide a list of corresponding responses, but rather redirect immediately
 491 to the `facebook.com` webpage. That is typically what *AllTheInter-*
 492 *net* is doing. This is treated in our stats as the SE providing a single
 493 response, ranked first. (An alternative approach would have been to
 494 assume a different q_1 for those cases—meaning a very high visibility
 495 for that page—but this would have altered the meaning of the tests we
 496 perform.)
- 497 • Sending too many requests over a short period results in our tool being
 498 banned by SEs like *Google* or *Startpage*; this is typically what happened
 499 after our campaign of tests presented in next section. We are working
 500 on preventing that behavior; a solution could be for the tool to run
 501 locally (on the user’s machine) instead of on our central server, which
 502 will reduce the searches per day from a given host; another way could
 503 be to have regulators intervene to legally prevent blocking when SEs
 504 are used for research purposes, as suggested in Europe.

¹¹For example, the urls `http://www.maps.com/FunFacts.aspx`, `http://www.maps.com/FunFacts.aspx?nav=FF`, `http://www.maps.com/FunFacts` actually lead to the same webpage.

- Sometimes, SEs update their platforms so that our parser may not get the relevant information to recover the top 10 results. We need to frequently update our tool to avoid those issues.

6. Comparisons on a large campaign of search terms

We now aim to analyze the results obtained from a campaign of searches. We chose to use 767 search terms among the most searched provided by Mondovo [39], and for which we also have the search volume, so that we can compute p_k for any search term $k \in \mathcal{K}$, and the related expected values (of scores, of positive tests...). Note that we ran our tool on the 1000 most-searched terms according to the mentioned website, but we encountered some data collection issues for 233 of them. The list of searched terms, as well as their search frequencies and the top 10 result we collected from each SE are publicly available¹². Interestingly, the search terms with data collection issues are not spread independently of their search volume: out of the 19 most-searched terms, 15 led to data collection issues, and they represent 39% of the total search volume from the 1000 most-searched terms. However, we believe this does not affect the analysis we make in this paper, because those specific terms are generally typed in a search engine merely to access the service with that name, not to obtain information regarding the search term. For example the top-searched term “facebook” is mostly searched by people using the SE almost as a DNS service: they just want to access their Facebook page. The other 14 terms with issues among the top 19 are of the same type: “youtube”, “google”, “gmail”, “xxnx”, “xvideos”, “facebook+login”, “ pornos”, “xxx”, “google+translate”, “yahoo”, “ebay”, “google+maps”, “yahoomail”, and “instagram”. Excluding those 15 search terms from the total search volume, the results we managed to collect represent 91.7% of the (remaining) search volume.

We plot in Figure 4 the volumes of searches for the 767 search terms in the study, as given by the aforementioned webpage. When plotted in double-logarithmic scale, the curve is close to a straight line, in accordance with the Zipf distribution, a distribution often observed in natural language, content or webpage popularity, and many other domains [40].

¹²All the data we use in this paper can be accessed in the folder <https://partage.imt.fr/index.php/s/6HWNPGit7BdAmXn>

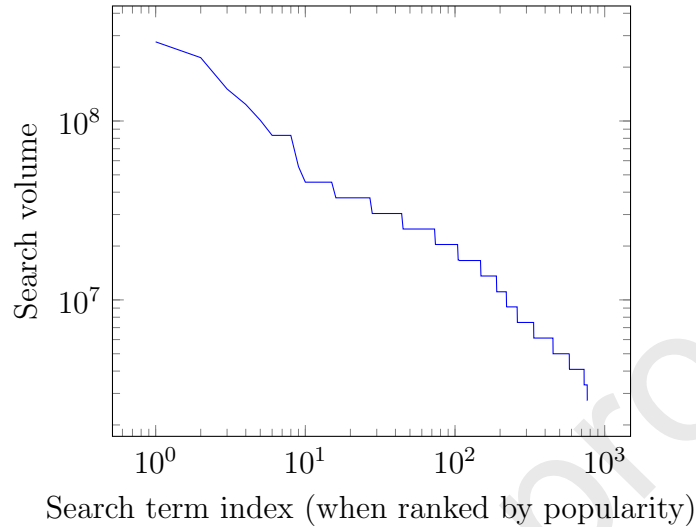


Figure 4: Search volumes of the 767 search terms used in the study

537 For the analysis in this section, we consider the 13 SEs given in the
 538 previous section, and the visibility/CTR values of Table 1.

539 6.1. Search engines scores

540 Table 2 gives the average (weighted) scores of the 13 search engines, plus
 541 the 2 meta engines. We also include the 95% confidence intervals that would
 542 be obtained (abusively) assuming requests are independently drawn from a
 543 distribution on all possible requests. One can check the score of her/his
 544 favorite SE.

545 To see how close an SE is with the consensus SE, Figure 5 displays by SE
 546 the percentage of common results with the consensus SE for each position-
 547 range. More specifically, for each SE we plot the proportion of links among
 548 the top x positions that are also in the top x positions of the consensus
 549 ranking.

550

551 Finally here, we draw in Figure 6 the distribution of the SE scores over
 552 searches (relatively to the consensus score), with on the x -axis the searches
 553 ordered (for each SE) by increasing relative scores. That representation al-
 554 lows to see how far each SE tends to be from the consensus ranking. It is
 555 not too surprising that the highest curve be the one from our other meta-SE

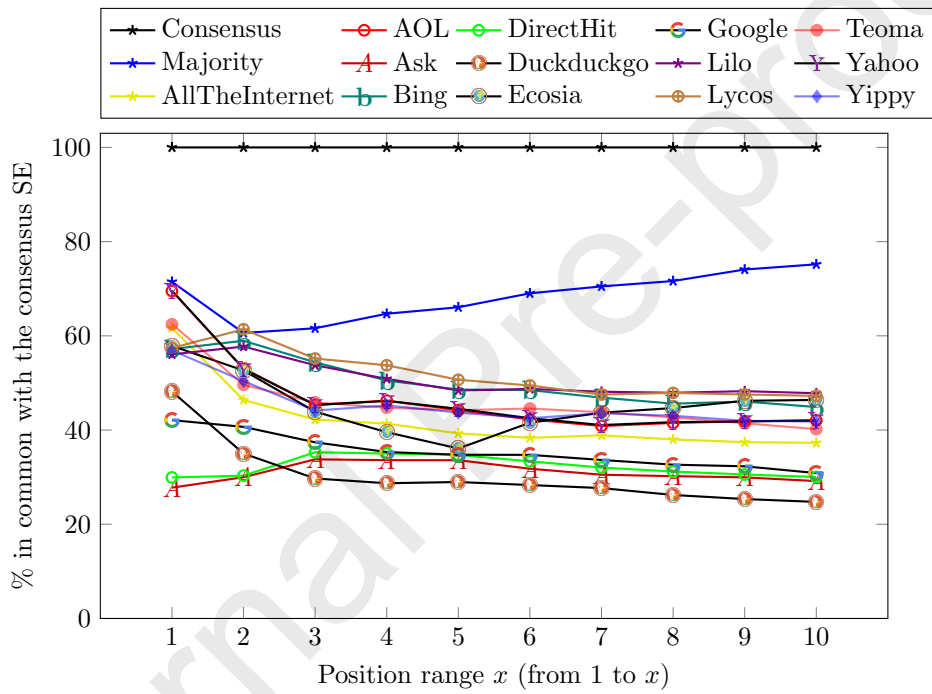


Figure 5: Similarities with the consensus within the first ranking positions.

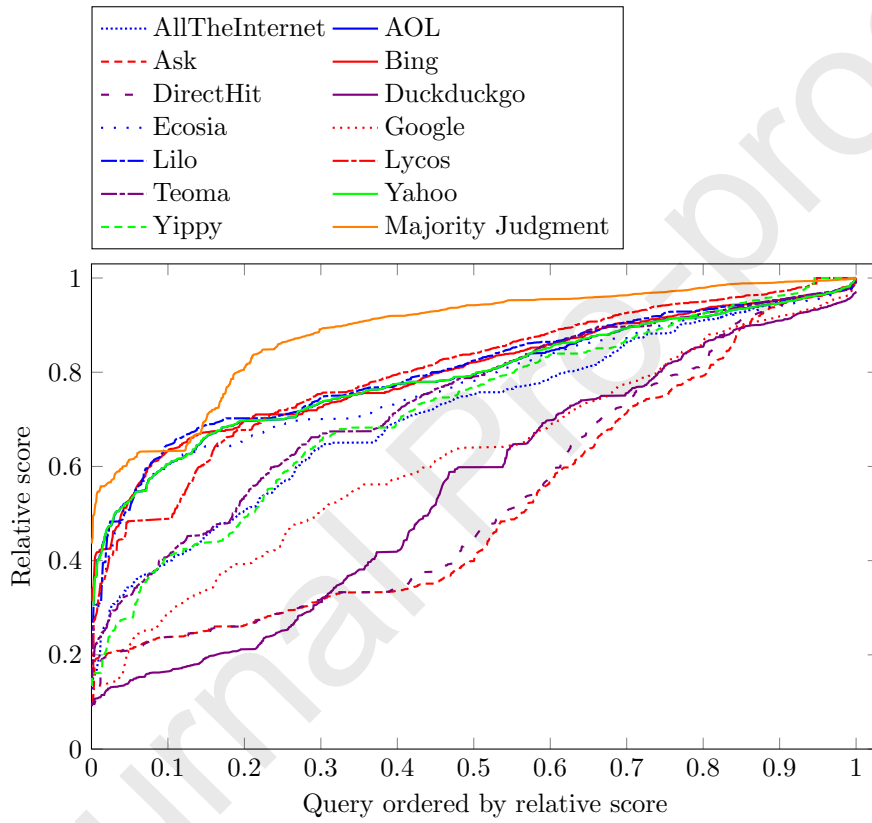


Figure 6: Distribution of scores relative to the consensus, from smallest to largest

SE	Score
Consensus	0.106 ± 0.002
Majority	0.096 ± 0.002
AllTheInternet	0.079 ± 0.003
AOL	0.087 ± 0.002
Ask	0.058 ± 0.003
Bing	0.087 ± 0.002
DirectHit	0.06 ± 0.003
Duckduckgo	0.062 ± 0.003
Ecosia	0.084 ± 0.002
Google	0.069 ± 0.003
Lilo	0.087 ± 0.002
Lycos	0.088 ± 0.002
Teoma	0.081 ± 0.003
Yahoo	0.087 ± 0.002
Yippy	0.08 ± 0.003

Table 2: SE average scores, and 95% confidence intervals half-widths.

(Majority Judgment), although it is at some points slightly below the curves for Yippy, Lycos, Lilo or Bing. Among SEs, one can also notice some distributions that look very close, like those from Ask and DirectHit, or those from Yahoo and AOL that almost coincide, or Lilo and Bing. Those apparent similarities need to be taken with care though, since for each SE the search terms are reordered by increasing values of the relative score, so having similar distributions does not imply that the SEs return the same results.

To investigate further that question of similarities between SEs, we compute an average (weighted) distance between each pair of SEs. The distance metric that we use is inspired by the Wasserstein distance used in optimal transport problems [41]: considering that a ranking gives some visibility weight to some webpages, we compute the minimum effort to move “visibility mass” (from some pages to others) to obtain the visibilities given by a second ranking. Since there is no clear distance defined between two webpages, we consider that any two different webpages are at the same distance (say, 1), so that the total effort to convert one ranking into another is just the amount of visibility to displace. That amount will then be considered as the distance between the two rankings. Mathematically, the distance (for a

	Majority	AllTheInternet	AOL	Ask	Bing	DirectHit	Duckduckgo	Ecosia	Google	Lilo	Lycos	Teoma	Yahoo	Yippy
Consensus	0.3	0.55	0.49	0.7	0.49	0.69	0.67	0.52	0.64	0.49	0.49	0.52	0.49	0.54
Majority		0.56	0.46	0.77	0.5	0.75	0.69	0.55	0.67	0.48	0.49	0.53	0.46	0.54
AllTheInternet			0.63	0.85	0.81	0.5	0.7	0.81	0.8	0.81	0.81	0.44	0.63	0.67
AOL				0.9	0.72	0.8	0.73	0.75	0.87	0.71	0.7	0.69	0.03	0.38
Ask					0.8	0.44	0.9	0.8	0.56	0.8	0.8	0.76	0.9	0.9
Bing						0.9	0.84	0.42	0.66	0.28	0.32	0.75	0.72	0.76
DirectHit							0.84	0.89	0.87	0.9	0.89	0.65	0.8	0.81
Duckduckgo								0.86	0.85	0.85	0.83	0.7	0.73	0.77
Ecosia									0.71	0.34	0.44	0.75	0.75	0.76
Google										0.67	0.68	0.67	0.87	0.88
Lilo											0.34	0.76	0.71	0.73
Lycos												0.74	0.68	0.72
Teoma													0.69	0.73
Yahoo														0.38

Table 3: Average Wasserstein distances between SEs (darker cells indicate a higher similarity). The average distance value is 0.67, the median 0.71, and the standard deviation 0.17.

574 given search) between rankings π and $\tilde{\pi}$ (with $\pi(i)$ the position of page i in
575 ranking π , and similarly for $\tilde{\pi}$) is therefore expressed as

$$D(\pi, \tilde{\pi}) = \sum_{i \in \mathcal{L}} [q_{\pi i} - q_{\tilde{\pi} i}]^+,$$

576 where $x^+ = \max(0, x)$. Those pairwise distances are given in Table 3. This
577 metric allows to see which SEs tend to give similar results. In particular,
578 the distance between AOL and Yahoo is very small, which is consistent with
579 our observations of Figure 6. Other SEs with relatively small distances are
580 the triplet Bing, Lilo, Lycos (with pairwise distances below 0.35). On the
581 other hand, some other pairs like Ask/DirectHit appear less similar than
582 what could be seen in Figure 6, although their distance remains more than
583 one standard deviation below the average value among SE pairs. Finally,

we notice that our two meta-SEs, the Consensus and Majority Judgment rankings, even if quite close when compared to other measured distances, are yielding clearly different results, with an average distance over our search terms of 0.3 (i.e., on average one has to “move” 30% of the “visibility points” to get from one meta-ranking to the other).

6.2. Identifying potential biases

We now see how SEs perform with respect to the tests defined in Section 3, for our set of search terms. Figure 7 presents the proportions of search terms (weighted by their frequency) for which each SE was rejected by our various tests (based on the Dixon outlier detection test) with risk 1%:

- for having a low score (test “SE score”, with tested hypothesis “*the score of that SE is not an outlier*”),
- for not showing high enough the most consensual page (test “Slot of top-consensus page”, with tested hypothesis “*the visibility given by that SE to the top-consensus page is not an outlier*”),
- for displaying in their top slot a page that is not deemed relevant by the other SEs (test “Top-SE page vs others”, with tested hypothesis “*the visibility given by that SE to its top-ranked page is not an outlier (among all visibilities given by SEs for that page)*”),
- for displaying in their top slot a page much less relevant than the others (test “Top-SE page scores”, with tested hypothesis “*the score of that SE’s top-ranked page is not an outlier (among all SE top-ranked-page scores)*”).

We note that Ask is the SE performing worst in all 4 test categories, followed closely by DirectHit and Duckduckgo, with Google a bit behind.

Looking at those 4 metrics, the best-performing SEs appear to be AOL, Bing, Lilo and Yahoo. However, note that this is likely to be a consequence of those SEs returning quite similar results, as shown before in Table 3: since the tests are based on outlier detection, a biased behavior will not be detected if implemented by enough SEs. Hence again, those results do not show that AOL, Bing, Lilo and Yahoo are better than the other SEs, but rather that their behavior do not individually stand out when compared to the 12 others. [To deal with that issue of redundancy among SEs, a possibility](#)

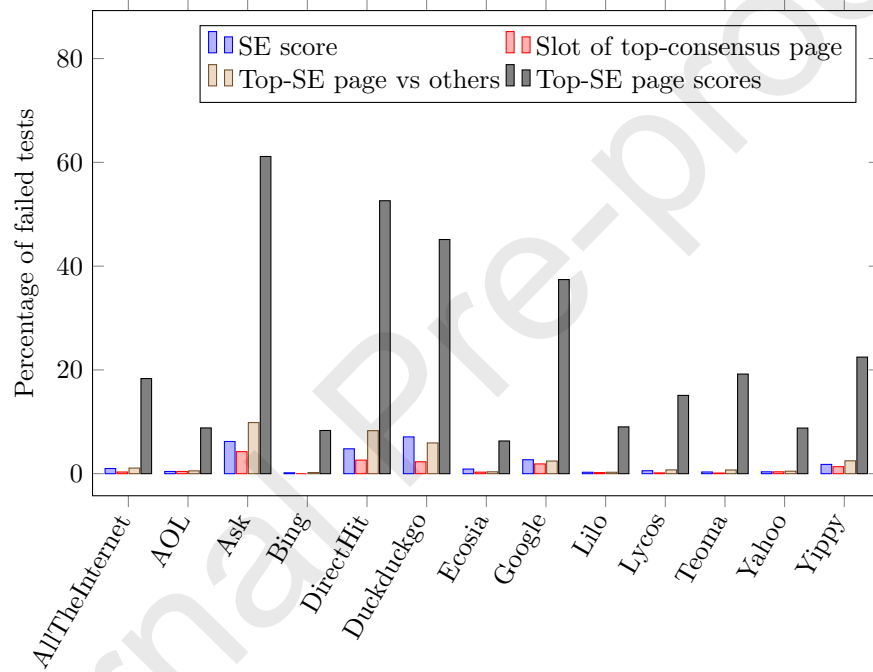


Figure 7: Proportion of failed tests (Dixon tests presented in Section 3) at risk level 1% for each SE (results for risk levels 5% and 10% are very similar).

would be to identify which SEs are redundant (e.g., through an appropriate statistical test) and then remove the redundancy by treating similar SEs as only one. For example, only one of each group of similar SEs could be kept while the other ones are ignored, but this would represent an asymmetrical treatment of those SEs; another option would be to consider the group as one SE, giving each webpage some visibility that is the average over the group (i.e., applying (1) but only with SEs in the group).

The fourth test experiences more detections, probably false positives, that test being more sensitive to the unsatisfied Gaussian assumption, comparing the (relatively much higher) CTR of the first position with potentially other CTR values if the page is not ranked first by other SEs.

To correlate with what we did in [32], we re-identify in Table 4 for each SE the (ordered) 10 queries with lowest relative score with respect to the consensus SE. The table seems to show less sensitive terms than what we

AllTheInternet	AOL	Ask	Bing	DirectHit	Duckduckgo	Ecosia
comuniato	focus	aright+move	coco	ebay+kleinanzeigen	ynet	mac
make+my+trip	paris	sears	zing	bbc+football	rambler	movistar
quot	allegro	9+gag	cars+game	american+online+inc	goggle+news	chromecast
bidmate	gossip	verizon+wireless	camera	sears	att	lazada
bee+bee	united+airlines	live+cricket+score	leos	9+gag	man+united	car+game
otto	eme	arsenal	michael	verizon+wireless	a+bola	zing
film+streaming	quot	rotten+tomatoes	captain+america+civil+war	live+cricket+score	citi+bank	tetris
camera+360	gaps	free+mobile	skies	arsenal	lowes	cheap+flights
mail+online	bonprix	lowes	happy+birthday	rotten+tomatoes	gamestop	odnoklassniki
bookers	fifa+16	national+hockey+league	fitbit	free+mobile	fandango	chatro
Google	Lilo	Lycos	Teoma	Yahoo	Yippy	
frozen	car+g	video+one	restaurant+near+me	focus	music	
restaurant+near+me	fahrenheit	anime+goog	pizza	paris	focus	
chrome	otto	east	tests	allegro	internet	
tesla	american+online+inc	coco	enterprise	gossip	cinopolis	
fitbit	emoji	mail+online	western+union	united+airlines	email	
currency+converter	fitbit	camera	traductor+de+google	mp3+music+downloading	east	
united+airlines	fifa+15	zing	avianca	eme	gossip	
pizza	chatro	happy+birthday	lego	quot	video+one	
mega	movistar	fitbit	app+lock	gaps	apps	
happy+wheels	happy+wheels	applock	cars+game	bonprix	nissan	

Table 4: 10 lowest-scores terms (relative to the consensus) for each SE.

had observed in [32], suggesting either that the discrepancies among SEs are not economically motivated or that such biases are not detected since implemented by several SEs. One can nevertheless notice the search term “chrome”, in the list of the terms for which Google’s score is the lowest relative to the consensus. For that term, one can expect that Google tends to refer more to its Chrome web browser rather than to the metal plating when compared with the other SEs; that specific search term is analyzed

further below.

Another observation which might be of interest is to look at the most important pages (that is, those having the largest visibility—their average CTR among SEs—over all weighted search terms) and compare their visibility at the various SEs. To do so, we aggregate pages within the same *domain*. The goal is to highlight SEs giving to some pages/domains a very different visibility from others: far above average could mean ties with the page, and far below would mean neglecting it in a suspect way. Figure 8 displays the distance with the average score (divided by the standard deviation), when we normalize the visibility scores. SEs are identified when at the extremes.

Figure 8 illustrates that Wikipedia is the most visible domain overall, even when its English-speaking and French-speaking variants are considered separate domains (all our searches have been performed from France). It seems that Google makes an extensive use of the geographical data of the user, and refers to the French-speaking version significantly more often than the other SEs; Google is also on the lower-end side for the visibility it gives to the English-speaking version. Finally regarding Google, one can notice that surprisingly, it is the SE giving the lowest visibility to YouTube, despite owning it. Maybe this is due to Google having a specific “video” option for its search, while all our tests were using the “all” (all types of results) filter.

The figure also highlights the higher visibility that Duckduckgo gives to Facebook and Twitter (visibility more than 3 standard-deviations above the SE average), and the low visibility it gives to some Google-owned services like Google Play, Google Translate, or just the Google website www.google.com. Notably, the SE Ecosia has a behavior opposite to Duckduckgo for those five domains.

6.3. Comparison of meta engines

We already illustrated in Section 4.2.2 in an artificial setting the advantage of the majority judgment ranking over the consensus one to reduce biases. Our goal here is to see what happens with “real” searches.

On Table 2, we can check that the majority judgment SE gives an average score about 10% smaller than the consensus one (built as the score-maximizing ranking). That score is significantly larger than any available engine, with the advantage of significantly reducing biases.

To emphasize the gain in bias reduction by the majority judgment ranking, we show in Figure 9, among the pages ranked first by an SE and identified

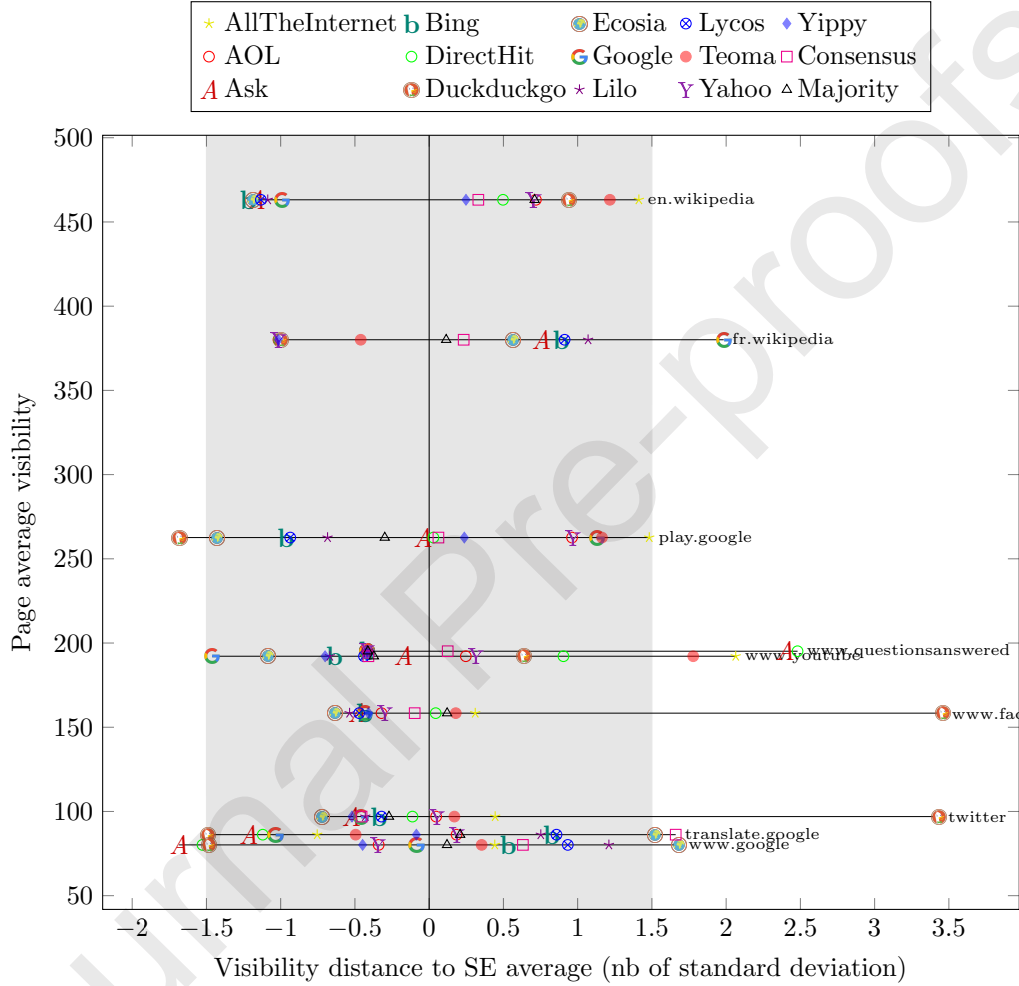


Figure 8: Distance between visibilities of domains for the most-shown domains. Each line is a domain, the line height is the domain average visibility, and each dot represents a specific SE (the abscissa of the dot is the difference between the domain visibility for that SE to the mean domain visibility, normalized with the standard deviation among SEs). The grey zone indicates the interval with semi-width 1.5 standard deviation.

674 by a Dixon’s Q test as a potential outlier (Section 3.4), their position distribution at both the consensus and majority judgment ranking. The figure

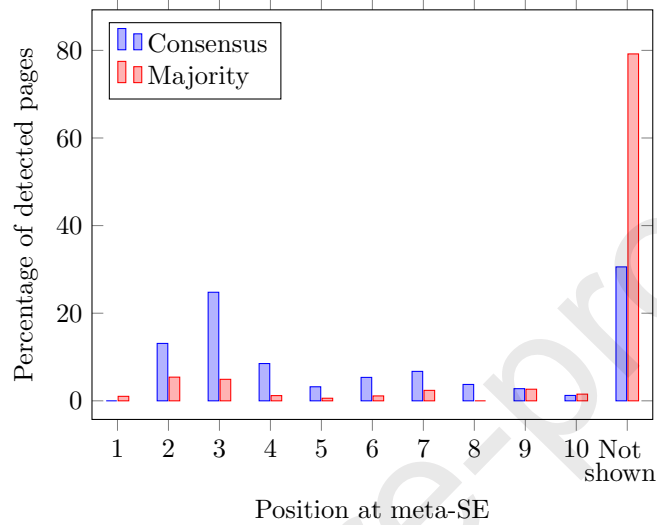


Figure 9: Repartitions of the position on the meta-SEs of the suspected top-results from SEs.

675 highlights the lower sensitivity of the Majority Judgment ranking to one page
 676 being artificially boosted by an SE, as we had observed for the numerical ex-
 677 periment of Section 4.2.2. Indeed, among the SE top-ranked pages detected
 678 as outliers due to a low score, the Majority Judgment ejects nearly 80%
 679 (*versus* 30% for the Consensus ranking), and generally tends to rank them
 680 worse than the Consensus ranking. That higher sensitivity of the Consensus
 681 ranking is due to the fact one SE showing a page on the top of its results
 682 already guarantees the page a score of at least q_1/N (attained if the page is
 683 only shown by this SE), which can be sufficient to be among the 10 highest-
 684 score pages; by contrast, the Majority Judgment would prefer a page that
 685 is shown—even in low positions—by two SEs or more over a top-result shown
 686 by only one SE.

688 6.4. An example: the “chrome” search term

689 We previously evoked the search term “chrome”, as one for which Google
 690 disagrees most with our Consensus meta-SE. In this section we analyze that
 691 case in more detail. The detailed results from each SE can be found on our

shared archive, to save space we only provide in Table 5 the rankings given by our two meta-SEs.

Consensus	Majority judgment
www.google.com/chrome	www.google.fr/chrome/index.html
www.google.fr/chrome/index.html	www.google.com/chrome/browser/index.html
www.google.com/chrome/browser/index.html	apps.apple.com/us/app/google-chrome/id53588..
www.questionsanswered.net/article/how-updat..	www.google.com/chrome
apps.apple.com/us/app/google-chrome/id53588..	support.google.com/chrome/answer/95346?co=G..
www.google.com/intl/en/chrome	play.google.com/store/apps/details?id=com.a..
www.commentcamarche.net/download/telecharge..	www.commentcamarche.net/download/telecharge..
www.clubic.com/telecharger-fiche257658-goog..	support.google.com/chrome/?hl=en
www.google.fr/chrome	www.clubic.com/telecharger-fiche257658-goog..
www.questionsanswered.net/article/how-insta..	chrome.google.com/webstore

Table 5: Rankings for our two meta-SEs for the research term 'chrome' (The two urls highlighted in red were shown only by two SEs, in their top two results).

Several points are worth mentioning for that specific search:

- The SEs Ask and DirectHit give the exact same three pages in their three top results, from the domain “questionsanswered.net”, but no page from that domain is shown by any other SE;
- Nevertheless, since there are two SEs showing the same top result, neither of them is labeled as an outlier regarding the visibility that their top-page has among others (test from Subsection 3.4), but both are detected by the other test version (from Subsection 3.5) comparing the visibility score of SE top-results.
- Those two SE, even if the only ones showing pages from the domain “questionsanswered.net”, influence significantly the Consensus ranking (due to the high weight given to top ranks) by getting two of those page shown. By contrast no such page is shown in the Majority Judgment ranking, illustrating its higher robustness to biases as observed for our synthetic example of Section 4.2.2.

7. Conclusions

This paper proposes a model and tools to analyze the results from search engines (SEs), combine them to build some meta-SEs, and detect possible bias using statistical tests. The whole framework is presented in a publicly-available tool. The basis of our model is the interpretation of the positions in search engines ranking in terms of *visibility* given to webpages. This allows us to aggregate the visibility that a webpage is given from a set of SEs, and to define several types of tests using the Dixon outlier detection test.

The meta-SEs we have defined and implemented have satisfying properties: the Consensus ranking allows to average out some possible bias ranks by ranking according to pages' aggregated visibility for a search term, and the Majority Judgment favors pages that deserve visibility by at least half of the SE, hence being more robust to manipulations by one or a few SEs.

The test campaign that we have carried out on 767 search terms allows to identify some differences and similarities among SEs, for example highlighting some tendencies to comparatively favor or disfavor specific domains.

The work presented in this paper opens several directions for future research. First, the data collection tool would need to be improved to be able to deal with SEs' changes in the presentation of their result pages, so that some further campaigns could be run on whole sets of search terms (for our campaign we had to remove 233 terms out of the initial 1000). Also, a fine tuning of the parameters in the model would be worth considering, in particular the visibility values for each position and on each SE would need to be estimated with more precision, and the results compared with what we get with our coarse approach (visibility depending only on position). We also compared results from "new" users, i.e., without any cookie, browsing or search history, or personal data; the method proposed in this paper could be used to investigate how such information affects SE results through test campaigns using a variety of user profiles, checking whether a bias occurs for some specific profiles.

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767 [unlocking_digital_competition_furman_review_web.pdf](https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/785547/unlocking_digital_competition_furman_review_web.pdf) (2019).
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